SCAN: Scattering Characteristics Analysis Network for Few-Shot Aircraft Classification in High-Resolution SAR Images

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***Abstract*— Recently, deep learning in synthetic aperture radar (SAR) automatic target recognition (ATR) has made significant progress, but the sample limitation problem in the SAR field is still obvious. Compared with the optical remote sensing images, the SAR images are insufficient, especially those containing the geospatial targets with certain target attitude angles (TAAs). To solve these problems, a novel few-shot learning framework named scattering characteristics analysis network (SCAN) is proposed in this article. First, a scattering extraction module (SEM) is designed to combine the target imaging mechanism with the network, which learns the number and distribution of the scattering points for each target type via explicit supervision. Besides, considering the imaging variability of SAR targets, a TAA-guided metalearning network consisting of an angle self-adaption classifier (ASC) and a frequency embedded module (FEM) is designed. ASC guides the network to focus on the positive sample pairs with different TAAs. FEM combines pulse cosine transform (PCT) with the network training process effectively to enrich frequency-domain information. In addition, a new dataset named SAR aircraft category dataset is constructed for the experiments. Compared with other few-shot SAR target classification approaches, our model efficiently integrates the scattering characteristics with the learning process, and the test accuracy for 5-way 1-shot has been improved by 4.74%. Finally, the experimental results are provided to demonstrate the validity of the proposed method.**

***Index Terms*— Aircraft, automatic target recognition (ATR), classification, deep learning, few-shot learning (FSL), scattering characteristics, synthetic aperture radar (SAR).**

I. Introduction

SYNTHETIC aperture radar (SAR) is an active imaging system with all-time and all-weather observing ability. Recently, with the rapid development of radar satellites, the demand for SAR image interpretation is increasing in disaster monitoring, topographic mapping, target detection and classification, and so on [1]. Particularly, SAR target classification has attracted more and more attention among researchers.

Over the years, the research on SAR target classification has made significant progress. There are roughly three ways: template-based, model-based, and machine learning-based. The template-based methods compare the test samples with the sample template of each class to classify the targets [2]. The model-based approaches describe the target characteristics with the help of physical models or conceptual models [3], [4]. For machine learning, several classifiers, such as AdaBoost and support vector machine (SVM), are designed for more accurate classification [5]–[7]. However, these approaches need a high cost in manual experience for both feature selection and classifier construction. Therefore, it is necessary to apply the convolutional neural network (CNN) to extract the target features comprehensively and automatically.

Benefitted from the success of deep learning in computer vision, deep learning has been introduced into the SAR domain [8]–[10] recently. However, deep learning models are data-driven, and the application of general deep learning methods in the SAR target classification task faces the challenge of insufficient samples for several reasons. First, limited by the way of data acquisition, the dataset in the remote sensing field is generally smaller than that in the natural scene. Compared with the optical remote sensing images, the SAR images are more insufficient [11]–[15], especially the images containing geospatial targets. Second, according to the SAR mechanism, the SAR target characteristics vary significantly with different target attitude angles (TAAs), as shown in Fig. 1. Generally, the geospatial targets in the SAR images are presented with certain TAA, which cannot meet the requirement of dataset diversity for deep learning. Furthermore, the rotation operation for data augmentation commonly used in optical images is not reasonable for SAR images. Third, the SAR target interpretation is a laborious task, which requires both expert experience and prior knowledge. Hence, the problem of lacking SAR samples with certain TAAs becomes more serious. In addition, deep learning algorithms are prone to overfitting in low data regimes, which declines the classification performance. Consequently, it is essential to apply the few-shot learning (FSL) to the SAR target classification tasks.

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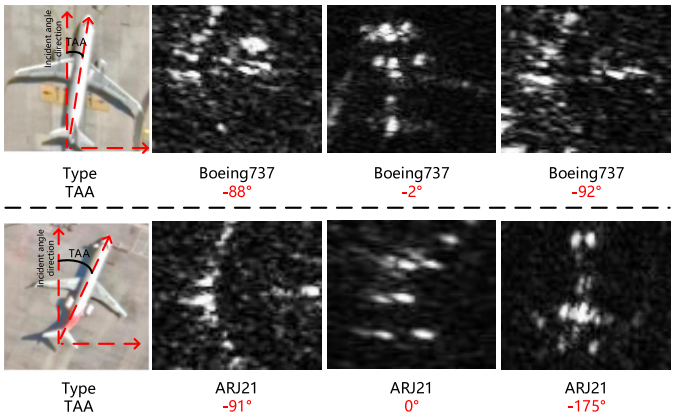


Fig. 1. Optical remote sensing images and the SAR images of the same category. The top is Boeing737, and the bottom is ARJ21. The TAA in this article is denoted as follows: The direction of the incident angle is denoted as 0°. It goes from 0° to −180° counterclockwise and from 0° to 180° clockwise.

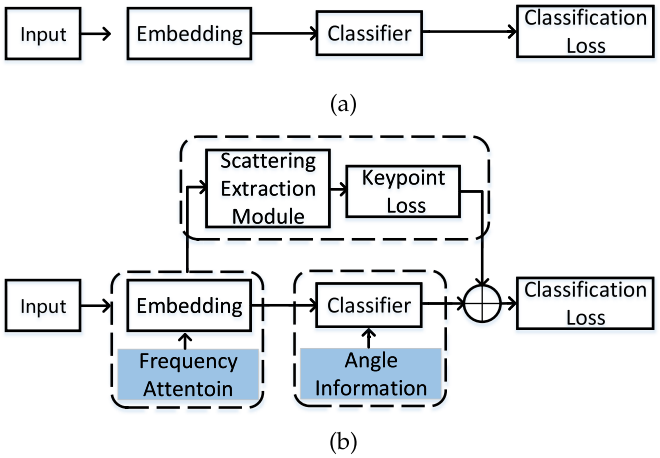


Fig. 2. Illustration of the FSL SAR target classification methods. (a) Previous methods are the general FSL methods in the natural scene. (b) SCAN (Ours) that combines scattering characteristics.

In the SAR domain, FSL methods roughly start from two directions: data expansion and model optimization [16]. Data expansion is a direct method to solve the sample limitation problem, including traditional data expansion methods, such as brightness and contrast conversion, and synthesis of new samples through simulation software or generative models [17]–[19]. However, data expansion methods for SAR images are complicated and tend to affect the classification efficiency. For model optimization, transfer learning and metalearning are two representative methods. Transfer learning aims to establish a connection between the source domain and the target domain, and maximizes the use of source-domain knowledge to solve the problem of insufficient samples in the target domain [19]–[28]. Metalearning is upgraded from sample-level learning to task-level learning, with the aim of achieving rapid learning of new tasks with only a few samples [29]–[36].

However, most previous model optimization studies directly apply the general models in natural scenes and do not consider the different imaging properties between SAR and optical images. Meanwhile, compared with the SAR ships, the discrete structure of the aircraft makes it more meaningful to take advantage of scattering characteristics. Generally, different from the optical imaging mechanism, the imaging result of SAR aircraft is composed of many discrete strong scattering points. The strong scattering points represent the maximum pixel value of the local scattering region in SAR images and have superior saliency and stability properties [37], [38]. Therefore, combining the strong scattering points is a feasible way to optimize the performance of SAR aircraft classification. Through the above analysis, a novel few-shot SAR target classification network named scattering characteristics analysis network (SCAN) is proposed. Different from previous methods, SCAN integrates the scattering characteristic and TAAs with the network learning process, as exhibited in Fig. 2. A scattering extraction module (SEM) serves as an auxiliary task to introduce additional supervision information to the network. Meanwhile, the target imaging result is sensitive to the TAAs, which makes the angular distribution of the aircraft sparse. Hence, a TAA-guided metalearning network is designed with the guidance of the TAAs. Compared with the few-shot SAR target classification approach [39] involving new categories, our method adopts the metric learning structure where fine-tuning is not required during the testing procedure.

The main contributions of our work can be summarized as follows.

1) We present a novel integrated framework named SCAN for few-shot SAR aircraft classification, which consists of a classification path and a scattering extraction branch. Our method shows the best performance on the novel category few-shot recognition tasks without fine-tuning.
2) To enhance SAR target feature representation, SEM is designed to assist in the optimization of the main classification task. It can guide the network to learn the number of strong scattering points and their spatial location distribution in different categories.
3) To alleviate the imaging variability of SAR targets caused by TAAs, a TAA-guided metalearning network consisting of a frequency embedded module (FEM) and an angle self-adaption classifier ASC) is proposed. FEM combines pulse cosine transform (PCT) with the training process effectively to enrich frequency-domain information. ASC guides the network to focus on the positive sample pairs with different TAAs during training and learn more representative and distinguishing features.

A new SAR aircraft classification dataset (SAR-ACD) is built for research. Experimental results on SAR-ACD show the superiority of SCAN. Compared with the baseline, the accuracies on the 5-way 1-shot and 5-way 5-shot tasks are improved by 4.74% and 3.18%, respectively.

II. Related Work

A. Data Expansion-Based Methods

For data expansion methods [40]–[44], they attempt to enlarge the dataset by generating new samples through

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adversarial learning or electromagnetic simulation. In [3], a new SAR target recognition method combining a computer-aided design (CAD) model with CNN is proposed. CAD models can flexibly generate images and features in different frequency bands and angles. The RaySAR method uses ray tracing technology to build simulation samples based on the 3-D model of the target [45]. Specifically, to solve the problem of lack of samples with specific angles, Song *et al.* [18] propose an adversarial autoencoder to generate SAR images with aspect angular diversity. Cui *et al.* [46] use Wasserstein generative adversarial network (WGAN) based on gradient penalty to generate new samples. They also propose a sample selection principle to select good samples with specific TAAs from the generated samples. Ding *et al.* [47] propose a sample synthesis method that generates new samples with a specific azimuth angle by performing a weighted summation of the two samples with the closest angle.

When we can guarantee the quality of the generated samples, data expansion-based methods are significantly effective and can solve the sample limitation problem fundamentally. However, most expansion methods are a little complex, and the flow is relatively slow.

B. Methods Based on Transfer Learning

Transfer learning methods [48]–[54] utilize the prior knowledge to alleviate the sample limitation problem in the target domain. The prior knowledge is learned from the source domain where the training data is sufficient. Reference [23] uses the optical cooperative institute for arctic research (CIFAR)-10 dataset to construct a pretrained network, and the middle feature layer is used for the classification of TerraSAR-X images. In [26], [53], the source data are replaced by simulated data. The transferred knowledge is learned from simulated data by constructing pretrained models. Specifically, to solve the problem of scarce samples with specific TAAs, [55] learns the mapping relation from sufficient data in the source domain, and the mapping can be applied to the target domain to generate targets with different aspect angles.

Generally, transfer learning has high requirements for source-domain data. The effect of transfer learning is closely related to the similarity between the source- and target-domain datasets.

C. SAR Target Classification Based on Metalearning

When there is only labeled data, metalearning methods are more appropriate. Especially, metalearning can achieve more stable and better recognition results when new categories are involved. When the number of samples is limited to a few shots, such as 1 or 5, only metalearning can recognize them effectively.

Metalearning aims to achieve rapid and effective learning when faced with a small sample of the new classes by learning a large number of old tasks. Metalearning methods can be divided into three categories: metric-based, model-based, and optimization-based [16]. Pan *et al.* [56] apply metric-based metalearning to solve SAR target classification problems through the Siamese network. The combination of the metric-based metalearning [57], [58] and the deep networks [59]–[61] is explored to achieve more effective deep embedding learning. Metatraining is implemented on the simulated dataset of moving and stationary target acquisition and recognition (MSTAR) to construct the general feature extractor. After that, the feature extractor parameters are fixed, and then, the rest network parameters are fine-tuned on the basis of the real MSTAR dataset in [62].

Specifically, to solve the problem of insufficient samples with specific TAAs, Zhao *et al.* [63] take multiaspect SAR target as input based on the metric-based framework, while, in most of the existing approaches, the images for each category in MSTAR are partitioned into the test set and the training set proportionally and do not involve new categories [34], [64]. Their purpose is to achieve higher accuracy by using fewer samples with the help of metatraining. Fu *et al.* [39] propose a metalearning framework that includes a metalearner and a base learner dedicated to learning a good initialization. Meanwhile, five categories in MSTAR are selected as the metatraining dataset, and the model's performance is tested on the remaining five new categories. More recently, Wang *et al.* [30] also apply metalearning to SAR target recognition and propose a hybrid inference strategy for classification on MSTAR. The metatraining dataset and the metatesting dataset contain seven and three categories, respectively.

III. METHOLOGY

The total structure of the proposed framework is shown in Fig. 3. The proposed network is mainly composed of three parts: ASC, SEM, and FEM. In this section, these three components will be described in detail. For simplicity, angle-guided (AG) classification loss is denoted as Loss_{ASC}, and the scattering extraction branch loss is denoted as Loss_{SEM}.

A. Problem Setup

Some metalearning concepts will be introduced first. One episode in episode training is similar to a batch in the traditional training method. The metalearning dataset is partitioned into the metatesting dataset and the metatraining dataset. The categories of the two dataset have no intersection. Assume that the metatraining dataset $D_{\text{train}} = \{(x_1, y_1), \ldots, (x_{N_{\text{tr}}}, y_{N_{\text{tr}}})\}$ contains a total of C_{tr} categories and N_{tr} samples. The metatesting dataset $D_{\text{test}} = \{(x_1, y_1), \ldots, (x_{N_{\text{te}}}, y_{N_{\text{te}}})\}$ contains C_{te} categories, and the number of test samples is N_{te}.

The metalearning process is shown in Fig. 4. In each training iteration, N categories are randomly selected from all training categories. Suppose that there are a total of T training categories; the number of different category combinations during training can be calculated as

$$C_T^N \cdot A_N^N = \left(\frac{T!}{N!(T-N)!}\right) \cdot (N(N-1)\cdots(1)). \quad (1)$$

Then, the support set and the query set are formed by randomly selecting K examples and Q examples from each selected category, respectively. That is, the model needs to learn how to classify these C types accurately from only N*K data, which is denoted as an N-way K-shot task. Similarly, in each testing iteration, the data formation is the same as the training process.

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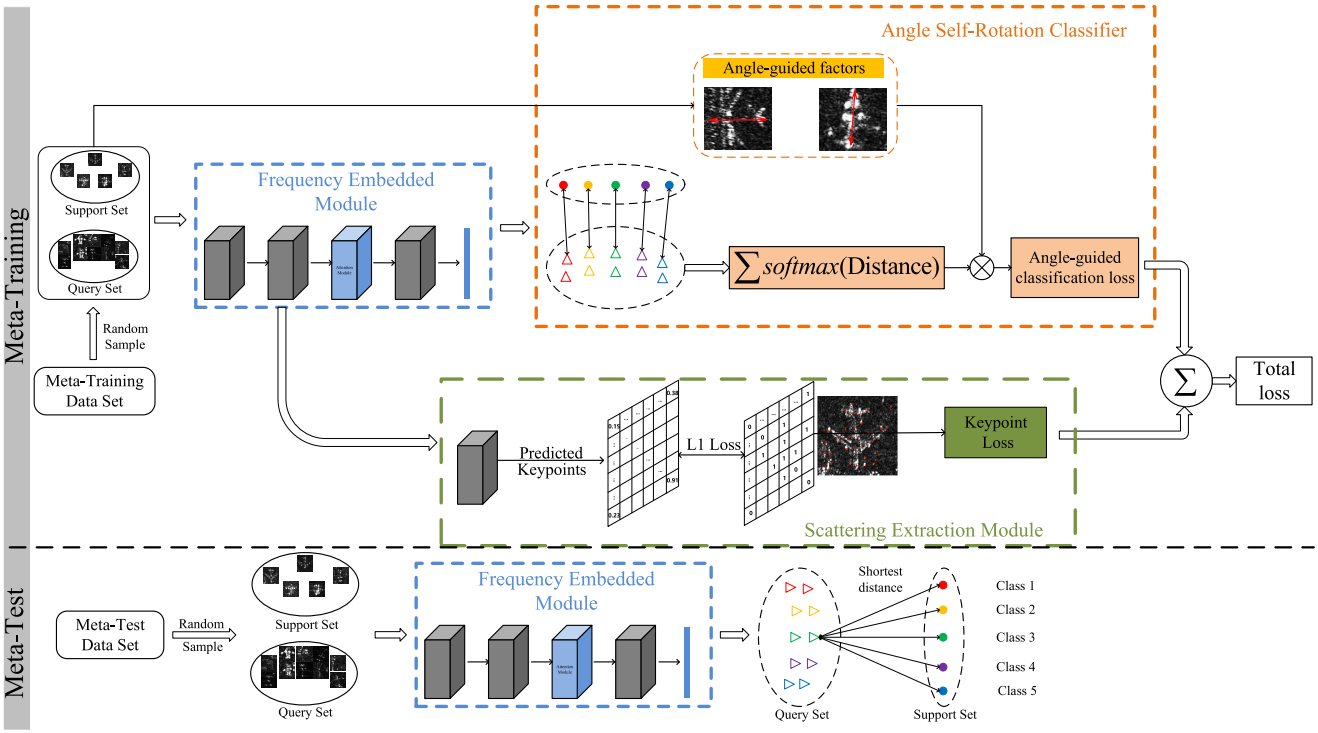


Fig. 3. Overall structure of SCAN. For the 5-way 5-shot case, AG factors are not used, and other modules are reserved. Both the SEM and the AG factors are not required in the testing phase.

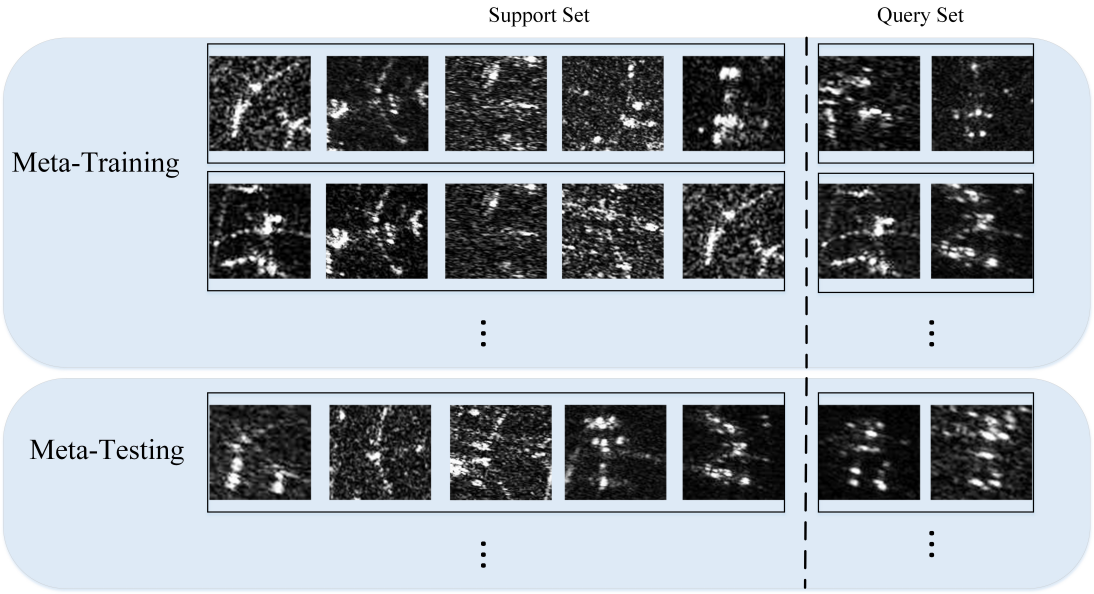


Fig. 4. 5way 1-shot setup. There is no intersection between the training categories and test categories. In each episode, one image is chosen from each of the five selected categories to form the support set, and 15 images are chosen from each of the five selected categories to form the query set.

Different metatasks will be randomly sampled during the training process in each training episode. This learning mechanism allows the model to learn the commonality of different metatasks and reduce focus on the specific task-related parts. As a result, the model's generalization ability is improved when faced with new metatasks.

B. Scattering Extraction Module (SEM)

In this section, a scattering extraction loss is proposed. A new convolution branch is added to extract the scattering keypoints of aircraft. In short, the network predicts the location and quantity information on keypoints simultaneously. The keypoints are extracted in the same way, as described in strong scattering point extraction.

The output of the second convolution block in the feature extraction module is served as the input of the scattering extraction branch. Generally, the keypoint prediction can be considered a pixel-level classification requiring us to decode the input feature map and resume it to a larger size. As a result, the deconvolution structure is introduced in this article

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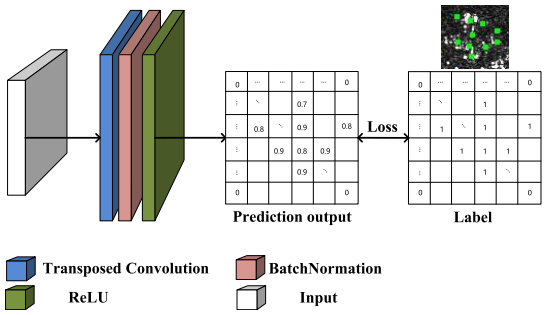


Fig. 5. Network structure of the SEM. Strong scattering points are preextracted by the Harris–Laplace detector and serve as supervision information.

to predict key points. The deconvolution process has learnable parameters, which can increase the network's fitting ability compared with up pooling and upsampling operations. The structure of the scattering extraction branch is shown in Fig. 5.

The coordinate conversion formula is used to calculate the position of the keypoint on the output. It is calculated as

$$y_2 = \frac{h_2}{h_1} \cdot y_1, \quad x_2 = \frac{w_2}{w_1} \cdot x_1 \tag{2}$$

where (w_1, h_1) and (w_2, h_2) illustrate the original image size and scaled size, respectively. (x_1, y_1) and (x_2, y_2) represent the original coordinates and the scaled coordinates, respectively.

Assume that the corresponding label key point map is $H \in \mathbb{R}^{50\times50}$, and one of the corresponding scaled coordinates of the strong scattering point is (x_2, y_2); then, $H(x_2, y_2) = 1$ and other areas are set to 0. The prediction key point feature map is $\hat{H} \in \mathbb{R}^{50\times50}$. The keypoint prediction loss is written as

$$\text{Loss}_{\text{SEM}} = \text{sigmoid}\left(\sum_{(i,j)} \hat{H}(i,j) - H(i,j)\right). \tag{3}$$

To encourage features representation learning via explicit supervision, the model parameters are restricted to update toward the set trajectory to generate more representative features.

C. Angle Self-Adaption Classifier (ASC)

1) Algorithm Description: In the few-shot SAR aircraft classification tasks, N labeled examples $S = (x_1, y_1), (x_2, y_2), \ldots, (x_N, y_N)$ of the support set are given, where each feature vector is $x_i \in R^D$ and the corresponding label is $y_i \in 1, \ldots, K$. The initial feature vectors are then mapped to M-dimension through an embedding function f_ϕ: $R^D \rightarrow R^M$ with learnable parameters ϕ [65]. Center representation vectors c_k are computed as follows:

$$c_k = \frac{1}{|S_k|} \sum_{(x_i, y_i)\in S_k} f_\phi(x_i) \tag{4}$$

where S_k denotes the sample set containing samples labeled with class k.

Given a distance function Dis: $R^M \times R^M \rightarrow [0, +\infty)$, the distances from query samples to the center representation vectors in the mapping space can be calculated [65]. The original distribution without the angle information is shown in 5

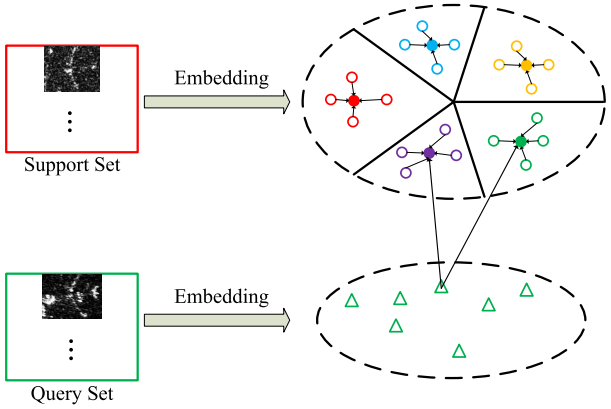


Fig. 6. Classification process for query images. The center representation vector of each class is got by calculating the mean vectors of the support set. The query vectors are classified according to the distance between them and the center representation vectors.

$$p_\phi(y = k|x) = \frac{\exp\left(-\text{Dis}\left(f_\phi(x), c_k\right)\right)}{\sum_{k'} \exp\left(-\text{Dis}\left(f_\phi(x), c_k\right)\right)}. \tag{5}$$

There are many distance metric functions, such as the Euclidean distance, the Manhattan distance, the Chebyshev distance, and the Minkowski distance. Experiments show that using the squared Euclidean distance is better.

SAR target imaging result is angle-sensitive. For the same type of aircraft, when the incident angle of the SAR image is the same or very close, the imaging results show significant differences with the increase in the TAA difference. Therefore, the ASC is designed to make the network more concerned with the positive sample pairs hard to distinguish. Naturally, the network will learn more representative and distinguishing feature representations.

2) Strong Scattering Point Extraction: The aircraft is basically composed of subcomponents, such as the wings, the tail, the fuselage, the nose, and the engines. It is found during the experiment that the Harris–Laplace detector shows better robustness and scattering point detection accuracy. Hence, the Harris–Laplace detector is selected to extract strong scattering points in this article. Strong scattering point detection results for different types of aircraft are given in Fig. 7. For different sizes and types of aircraft, such as Boeing and Airbus, it verifies the rationality of the distribution of the extracted scattering points.

3) Acquisition of the TAAs by the Least-Squares Method: It is laborious to obtain the corresponding TAAs by manual annotation for an aircraft instance. A novel method for calculating the TAAs is devised with the extracted strong scattering points. There are three types of typical SAR aircraft echo characteristics: one is the "V" type, with swept-back wings; the second is the "T" type, which is a flat-wing aircraft; the third is the double dot or "peanut" type, which is the helicopter. Considering the SAR aircraft can be regarded as a series of strong scattering points and the symmetry of the

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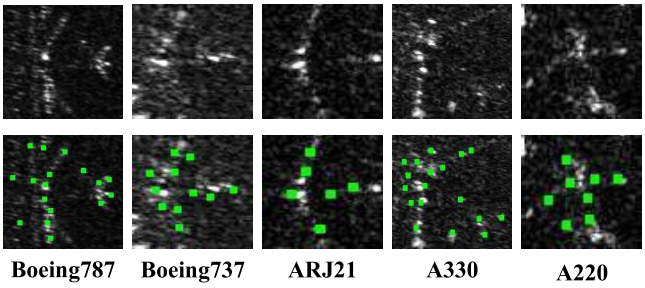


Fig. 7. Strong scattering point extraction results of different types of aircraft. Initial images are shown in the upper line, and detected images are shown in the bottom line.

above three structures, it is reasonable to use the least-squares method for linear regression on these strong scattering points. The slope of the line obtained by regression corresponds to the TAA. As shown in Fig. 7, assume that n points are extracted by the Harris–Laplace detector and define these points as $[(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)]$. We then calculate the respective average values of the abscissa and ordinate of them

$$\bar{x} = \frac{x_1 + x_2 + \cdots + x_n}{n}$$
$$\bar{y} = \frac{y_1 + y_2 + \cdots + y_n}{n}. \tag{6}$$

Then, the intercept and slope of linear equation can be calculated according to the average value above

$$b = \frac{n\sum_{i=1}^{n} x_i y_i - \sum_{i=1}^{n} x_i \sum_{i=1}^{n} y_i}{n\sum_{i=1}^{n} x_i^2 - \left(\sum_{i=1}^{n} x_i\right)^2} \tag{7}$$
$$k = \bar{y} - b\bar{x} \tag{8}$$

where b and k are the intercept and slope of the linear equation, respectively. The final angle is

$$\theta = \arctan(k). \tag{9}$$

4) Angle-Guided Classification Loss Function: For the 5-way 1-shot tasks, the support set S contains $5*1 = 5$ images. Suppose that there are N_Q examples in the query set Q for each episode. The original classification loss is denoted as

$$\text{Loss} = -\text{logsoftmax}\left(\sum_{q\in Q} \text{sqrt}\left(\text{sum}\left(\left(q - s_{y_q}\right)^2\right)\right)\right) \tag{10}$$

where $q \in Q$ and $s \in S$ are the embedding vectors obtained from the feature extraction module. y_q is the corresponding category label of q, and s_{y_q} is the support vector of the corresponding category label of q. The AG classification loss combined can be written as

$$\text{Loss}_{\text{ASC}} = -\text{logsoftmax} \times \left(\sum_{q\in Q} f\left(\theta_q - \theta_{s_{y_q}}\right)\text{sqrt}\left(\text{sum}\left(\left(q - s_{y_q}\right)^2\right)\right)\right) \tag{11}$$

where θ_q and $\theta_{s_{y_q}}$ are the TAA of q and s_{y_q}, respectively. $f(\cdot)$ is denoted as

$$f(x) = \text{softmax}(\text{abs}(x))$$
$$x = \left[\theta_q - \theta_{s_{y_q}}\right], \quad q \in Q. \tag{12}$$

In this situation, the network will focus more on the positive sample pairs with the same category but have large differences in the distribution of strong scattering points.

The total loss is

$$J = \text{Loss}_{\text{ASC}} + \frac{1}{N_Q}\text{Loss}_{\text{SEM}} \tag{13}$$

by minimizing the loss of the query samples at each episode via Adam to achieve metalearning.

D. Frequency Embedded Module (FEM)

Frequency-domain learning has always been a powerful measure for image processing. The frequency-domain analysis should be more meaningful for SAR target interpretation considering the special imaging mechanism of SAR images.

1) Basic Frequency Channel Attention Network: Assume that the input image is $I \in \mathbb{R}^{3\times A\times D}$, $X \in \mathbb{R}^{C'\times A'\times D'}$ is the output feature map, and D' and A' are the width and the height of the feature map, respectively. C' is the number of channels. Different channels represent the image components on different convolution kernels. Similar to time–frequency transformation, the convolution operation is similar to the Fourier transform. Thus, the information of one channel can be decomposed into signal components on C convolution kernels [66]. Typically, the 2-D discrete cosine transform (DCT) is formulated as

$$f_{a,d}^{2d} = \sum_{i=0}^{A-1}\sum_{j=0}^{D-1} x_{i,j}^{2d}\cos\left(\frac{\pi a}{A}\left(i + \frac{1}{2}\right)\right)\cos\left(\frac{\pi d}{D}\left(j + \frac{1}{2}\right)\right)$$
$$\text{s.t. } a \in 0, 1, \ldots, A-1, \quad d \in 0, 1, \ldots, D-1 \tag{14}$$

where $f^{2d} \in \mathbb{R}^{A\times D}$ is the 2-D DCT frequency spectrum, $x^{2d} \in \mathbb{R}^{A\times D}$ is the input, A is the height of x^{2d}, and D is the width of x^{2d}.

In [66], it has been proven that the other frequency components and information are discarded in the global average pooling except for the lowest frequency component. The larger coefficient of dimensional DCT indicates that the visual features of the corresponding frequency and direction appear more frequently in space.

2) PCT: The PCT model suppresses the visual features that appear more frequently in the visual space by whitening (normalized) the amplitude values of the DCT coefficients. It has been proven effective for SAR image interpretation [67], [68].

After obtaining the frequency-domain characteristics through the 2-D DCT, the transformed output will be quantized into 1, 0, and -1 in the frequency domain. This operation imitates the biological visual mechanism. 1 and -1 transformations, respectively, indicate the activation and deactivation of the signal at the corresponding position, and 0 transformation means that the information is inhibited.

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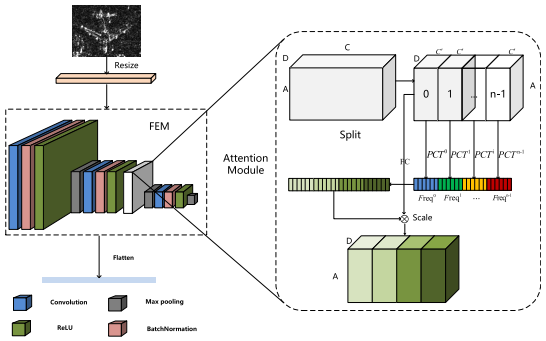


Fig. 8. Structure of the FEM. The resized images are sent to the embedding network to obtain the flattened feature vectors.

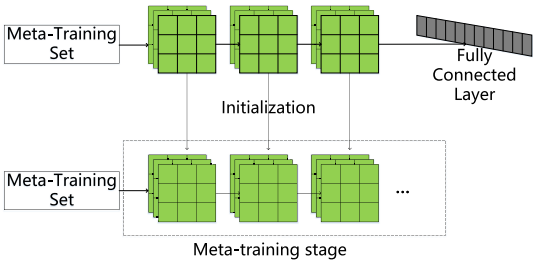


Fig. 9. Metalearning process combined with pretraining.

The quantization process of the transformed feature map in the frequency domain can be expressed as

$$I_n(u_i, v_i) = \text{sign}(I_o(u_i, v_i)) \begin{cases} -1, & \text{if } I_o(u_i, v_i) < 0 \\ 0, & \text{if } I_o(u_i, v_i) = 0 \\ 1, & \text{if } I_o(u_i, v_i) > 0 \end{cases} \quad (15)$$

where (u_i, v_i) represents the coordinates of pixel i, and $I_o(u_i, v_i)$ and $I_n(u_i, v_i)$ represent the original value and the transformed value of pixel i, respectively. Besides, sign$(\cdot)$ represents the symbolic function, the positive DCT coefficient in the feature map is quantized to $+1$, the negative DCT coefficient is quantized to -1, and the original data are retained when the DCT coefficient is 0. Fig. 8 displays the architecture of the frequency embedded network. The feature map is partitioned into n parts in channel dimension, and each part corresponds to different parameters.

E. Combination With the Pretraining Phase

Due to the limitation of SAR data, it is meaningful to take full advantage of the existing data. Therefore, a pretraining phase is introduced to get more prior knowledge and better model initialization parameters. As described in Fig. 4 for metalearning, the metalearning dataset consists of the training and test sets. The pretraining phase is implemented on the training set before the metatraining stage, and the details are displayed in Fig. 9.

IV. Experiment

In this section, we will perform different groups of experiments to verify the effectiveness of the metalearning principle and the SCAN. According to the dataset partition form, we conduct two groups of experiments.

In the first group of an experiment, the metatest dataset and the metatraining dataset are made up of six and 14 categories, respectively. For the second group of the experiment, all 20 categories are used in both the training and test sets. The images of each class are partitioned into the test set and the training set based on the ratio of 2:8. The model adopts the 20-way 1-shot metatraining module. To simplify, the two groups of experiments are named **Experiment I** and **Experiment II**, respectively.

A. Dataset Description

1) Data Sources: Similar to Gaofen (GF)3-aircraft detection data set (ADD) in [37] and FUSAR-Ship in [69] where GF-3 C-band images are utilized for SAR ship recognition, an experimental dataset named SAR-ACD is constructed to illustrate the effectiveness of the proposed network. A total of 11 GF-3 C-band images with 1-m resolution working in the Spotlight mode are used in this experiment. These GF-3 satellite images cover the Shanghai Hongqiao Airport area, Beijing Capital International Airport area, and another airport in different time phases with horizontal–horizontal (HH) polarization. The dataset contains six civil aircraft categories and 14 other aircraft categories. A total of 4322 aircraft clips are contained in SAR-ACD. It should be emphasized that SAR-ACD contains complex scenes at different airports. Meanwhile, the targets in SAR-ACD are rich in categories and have large size differences. These factors lead to a very challenging classification task.

2) Category Labeling: These SAR data are manually labeled according to the corresponding optical images. Generally, the aircraft stand is fixed. The ground truth of aircraft can be considered reliable and accurate by combining expert interpretation and large manual annotation.

After collecting the initial SAR images, normalization and radiation calibration are implemented to correct the images.

Civil SAR aircraft and the corresponding optical images are shown in Fig. 10. The specific categories of civil aircraft and the target number of each category are shown in Fig. 11. Several data expansion methods, such as contrast transformation, brightness transformation, and sharpness transformation, are adopted to address the issue of the imbalanced number of instances. As the imaging result of the SAR target varies greatly as the TAA changes, the operations such as rotation and reversal are not used.

B. Implementation Detail

1) Data Setting Details: In **Experiment I**, the ablation studies on the SEM and FEM before and after pretraining are performed, respectively. The experimental results also prove the significance of the pretraining phase. The pretrained model is obtained by the 14-class classification experiment on the metatraining dataset, which can fully utilize the prior information of the existing samples.

In addition, the metatraining stage is trained with supervised learning. The Adam optimizer is used for training where the learning rate is set to 0.001. The model is trained for 10 000 epochs, and if the verification results do not improve after 200 epochs, the training process will be terminated. Besides,

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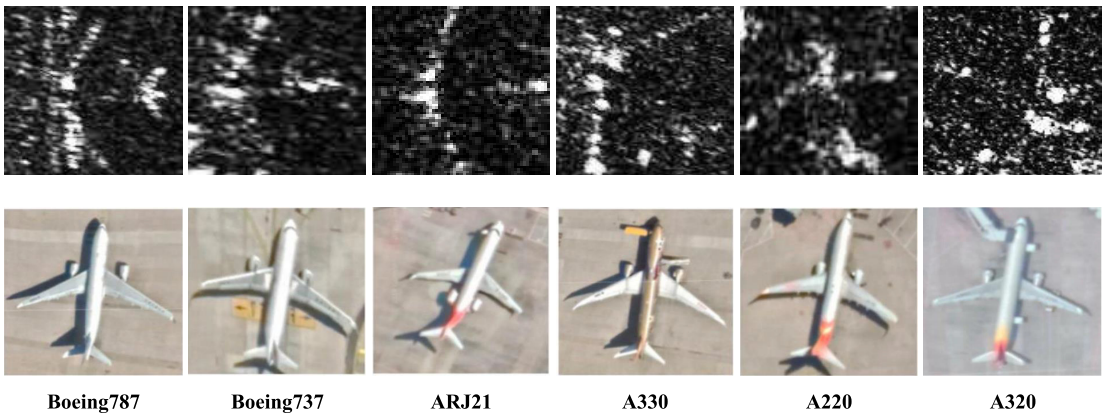


Fig. 10. Civil aircraft in SAR-ACD and the corresponding optical images.

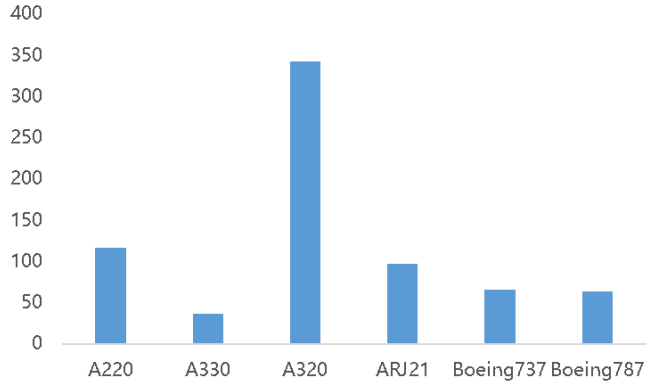


Fig. 11. Original number of each type of civil aircraft cropped from the original large-scale images without data expansion. The number of each category of aircraft after expansion is around 400.

100 train episodes and 100 test episodes are carried out per epoch. In the metatest stage, the best model obtained in the metatraining stage is tested on 1000 randomly selected N-way K-shot tasks. There are two types of tasks: 5-way 5-shot and 5-way 1-shot tasks. As shown in Table V, the number of query examples per class remains 15 for both of them.

In **Experiment II**, we change the proportion of training images from 10% to 100% to fully verify the effectiveness of the metalearning way. Similarly, the training mode and the optimizer setting are the same as **Experiment II**. The N-way K-shot tasks are sampled from the training set of all 20 categories. In **Experiment II**, the metatraining and metatest ways need special attention. Unlike **Experiment I** where the sampled tasks are 5-shot tasks or 1-shot tasks with five classes, the basic task unit in **Experiment II** is 20-way 5-shot and 20-way 1-shot. To ensure a fair comparison with classical fully connected networks (FCNs), the number of query images per class for metatraining and metatesting is 15 and 2, respectively.

2) Network Structure Details: The construction of the FEM is exhibited in Fig. 8. FEM consists of three convolutional blocks. For each convolutional block, it consists of a convolutional layer with 64 filters and 3 × 3 convolution, batch normalization (BN), rectified linear unit (ReLU) nonlinearity, and 2 × 2 max-pooling. The single-channel input images of different sizes are first resized to 1 × 100 × 100 uniformly, and the output sizes of the three convolution blocks of the feature extraction module are 64 × 25 × 25, 64 × 12 × 12, and 64 × 6 × 6, respectively. The output of the feature extraction module is then flattened and sent into the AG-classifier 6. For the SEM branch, the input size is 64 × 25 × 25, and the stride of the transposed convolution layer is set to 2. It is noticeable that the size of the SEM output is

TABLE I
ABLATION STUDY OF FEM AND SEM (**EXPERIMENT I**)

SEM	FEM	5-way 1-shot(%)	5-way 5-shot(%)
×	×	53.18 ± 0.62	68.66 ± 0.51
×	✓	53.54 ± 0.63	69.9 ± 0.50
✓	×	53.57 ± 0.62	70.04 ± 0.52
✓	✓	**55.9± 0.64**	**71.04± 0.52**

TABLE II
CLASSIFICATION ACCURACY COMBINED WITH PRETRAINING ON THE 5-WAY 1-SHOT CONDITION (**EXPERIMENT I**)

SEM	FEM	No pre-training(%)	Pre-training(%)
×	×	53.18 ± 0.62	55.21 ± 0.63
×	✓	53.54 ± 0.63	55.28 ± 0.64
✓	×	53.57 ± 0.62	55.55 ± 0.64
✓	✓	**55.9± 0.64**	**56.05± 0.65**

TABLE III
CLASSIFICATION ACCURACY COMBINED WITH PRETRAINING ON THE 5-WAY 5-SHOT CONDITION (**EXPERIMENT I**)

SEM	FEM	No pre-training(%)	Pre-training(%)
×	×	68.66 ± 0.51	69.04 ± 0.52
×	✓	69.9 ± 0.50	71.41 ± 0.50
✓	×	70.04 ± 0.52	70.50 ± 0.50
✓	✓	**71.04± 0.52**	**71.84± 0.50**

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TABLE IV
COMPARATIVE EXPERIMENT ON AG-CLASSIFIER WITH THE COMBINATION OF PRETRAINING (**EXPERIMENT I**)

FEM	SEM	No ASC(%)	With ASC(%)
×	×	55.21 ± 0.64	57.01 ± 0.64
✓	×	55.28 ± 0.63	57.82 ± 0.63
×	✓	55.55 ± 0.64	57.85 ± 0.62
✓	✓	56.05 ± 0.65	$\mathbf{57.92 \pm 0.60}$

TABLE V
DETAILS OF DATA SETUP DURING TRAINING AND TESTING (**EXPERIMENT I**)

Setup	5-way 5-shot		5-way 1-shot	
	support set	query set	support set	query set
meta-training	5	15	1	15
meta-testing	5	15	1	15

TABLE VI
DETAILED SETTINGS FOR METATRAINING AND METATESTING IN THE 20-CLASS CLASSIFICATION EVALUATION (**EXPERIMENT II**)

Setting	Support Set	Query Set
meta-training	20*1=20	20*15=300
meta-testing	20*1=20	20*2=40

$1 \times 50 \times 50$, while the keypoints are extracted on the initial input images with varying sizes. For the value in the keypoint feature map, 1 denotes that the corresponding position is a keypoint, and 0 denotes that the corresponding position is noncritical components or background.

The details of the metatraining and metatest process are shown in Algorithms 1 and 2, respectively.

Algorithm 1 SCAN for Metatraining

Input:Meta-training data set $D_{\text{train}} = \{(x_1, y_1), \ldots, (x_{N_{\text{tr}}}, y_{N_{\text{tr}}})\}$, where each $y_i \in \{1, \ldots, C_{\text{tr}}\}$. D_k contains all samples belonging to class k in D_{train}.
for *each episode* **do**
  RANDOMSAMPLE($\{1, \ldots, C_{tr}\}$,N)
  for k *in* $\{1, \ldots, N\}$ **do**
    RANDOMSAMPLE(D_k,K) $\rightarrow S_k$
    RANDOMSAMPLE(D_k,Q) $\rightarrow Q_k$
    Compute center representation vector c_k following Equation 4.
  end
  for k *in* $\{1, \ldots, N\}$ **do**
    for $(x_q, y_q), (x_s, y_s)$ *in* Q_k, S_k **do**
      Compute the TAAs of x_q and x_s according to Equations. 6-9.
    end
    Compute $Loss_{angle}$ according to Equation 11.
    Compute $Loss_{pre}$ according to Equation 3.
    Compute total loss J following Equation 13.
  end
end

Algorithm 2 SCAN for Metatesting

****** **Meta test** ******
for *each episode* **do**
  RANDOMSAMPLE($\{1, \ldots, C_{te}\}$,N)
  for k *in* $\{1, \ldots, N\}$ **do**
    RANDOMSAMPLE(D_k,K) $\rightarrow S_k$
    RANDOMSAMPLE(D_k,Q) $\rightarrow Q_k$
    Compute center representation vector c_k following Equation 4.
  end
  for (x_q, y_q) *in* Q_k **do**
    for k *in* $\{1, \ldots, N\}$ **do**
      Compute Distance(x_q, c_k)
    end
    Result:$x_q \in CLASS$ $\arg\min_k(Distance(x_q, c_k))$
  end
end

C. *Experiment I* on SEM and FEM

This section performs the ablation study with and without pretraining on the two modules to verify their effectiveness. The data setup for the metatraining and metatest stage is shown in Table V.

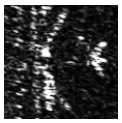
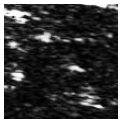
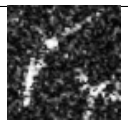
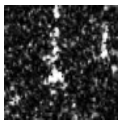
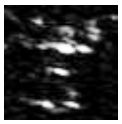
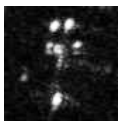
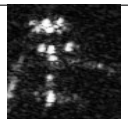
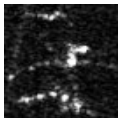
1) Ablation Study: As shown in Table I, the model shows better performance with the help of the two modules. When only with FEM, the test accuracy increased by 0.4% (1-shot) and 1.3% (5-shot), respectively. When only with SEM, the test accuracy is increased by 0.4% (1-shot) and 1.4% (5-shot), respectively. For 1-shot, the classification accuracy even increased by 2.8% with both modules. The ablation results fully verify the effectiveness of the two modules.

Besides, all channels of the last convolution layer are visualized to understand what the network has learned with SEM intuitively. As shown in Fig. 12, there are 64 channels in the last convolutional layer. The network focuses more on the critical parts of the aircraft and less on the edge and background noise.

2) Interpretability Analysis for Application on SAR Data: As shown in Fig. 13, the two shared convolution layers of the network with SEM and FEM are visualized. The top row is A220, and the bottom row is Boeing787. After adding SEM and FEM, the network pays more attention to the key parts, such as the nose, wings, and tail. Meanwhile, the feature maps of different aircraft based on different models are also visualized. Fig. 14(b) and (c) represents the visualization of the first convolutional layer of the baseline model and the model with SEM and FEM, respectively. Fig. 14(d) and (e) represents the visualization of the second convolutional layer

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TABLE VII
TEST RESULTS ON SOME AIRCRAFT PATCHES OF THE METATEST SET (**EXPERIMENT I**)

N-way K-shot	TAAs	Images	Ground Truth	baseline	With only SEM	With only ASC	SCAN
5-way 1-shot	−92°		Boeing787	ARJ21 ×	Boeing787 ✓	Boeing787 ✓	Boeing787 ✓
	−41°		Boeing787	A220 ×	Boeing737 ×	A220 ×	Boeing787 ✓
	−45°		A220	Boeing787 ×	A220 ✓	A220 ✓	A220 ✓
	−122°		A220	Boeing787 ×	A220 ✓	A220 ✓	A220 ✓
5-way 5-shot	5°		ARJ21	A220 ×	Boeing737 ×	Boeing737 ×	A220 ×
	8°		ARJ21	Boeing737 ×	Boeing737 ×	ARJ21 ✓	ARJ21 ✓
	0°		Boeing737	ARJ21 ×	Boeing737 ✓	Boeing737 ✓	Boeing737 ✓
	42°		Boeing737	A220 ×	ARJ21 ×	Boeing737 ✓	Boeing737 ✓

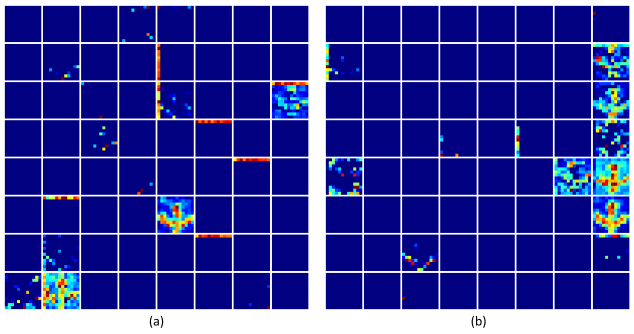


Fig. 12. Visualization of all channels in the last convolution layer (**Experiment I**). (a) Baseline network. (b) Network with SEM.

TABLE VIII
CLASSIFICATION ACCURACY WITH DIFFERENT PROPORTIONS OF THE TRAINING SET ON THE 20-WAY 1-SHOT CONDITION (**EXPERIMENT II**)

Percent	FCN(%)	baseline(%)	With ASC(%)
10%	32.09	69.18 ± 0.26	75.31 ± 0.26
30%	42.98	81.24 ± 0.27	89.32 ± 0.26
50%	56.75	93.34 ± 0.27	94.47 ± 0.27
70%	58.36	95.30 ± 0.27	96.01 ± 0.27
100%	62.33	95.75 ± 0.26	97.22 ± 0.27

of the baseline model and the model with SEM and FEM, respectively. Red represents places with high network attention. From the visualization of the two shared convolutional layers, it can be seen that the network's attention to the background decreases, and the attention to the aircraft tail increases.

3) Combination With the Pretraining Phase: Beyond that, a comparison experiment with or without the pretraining

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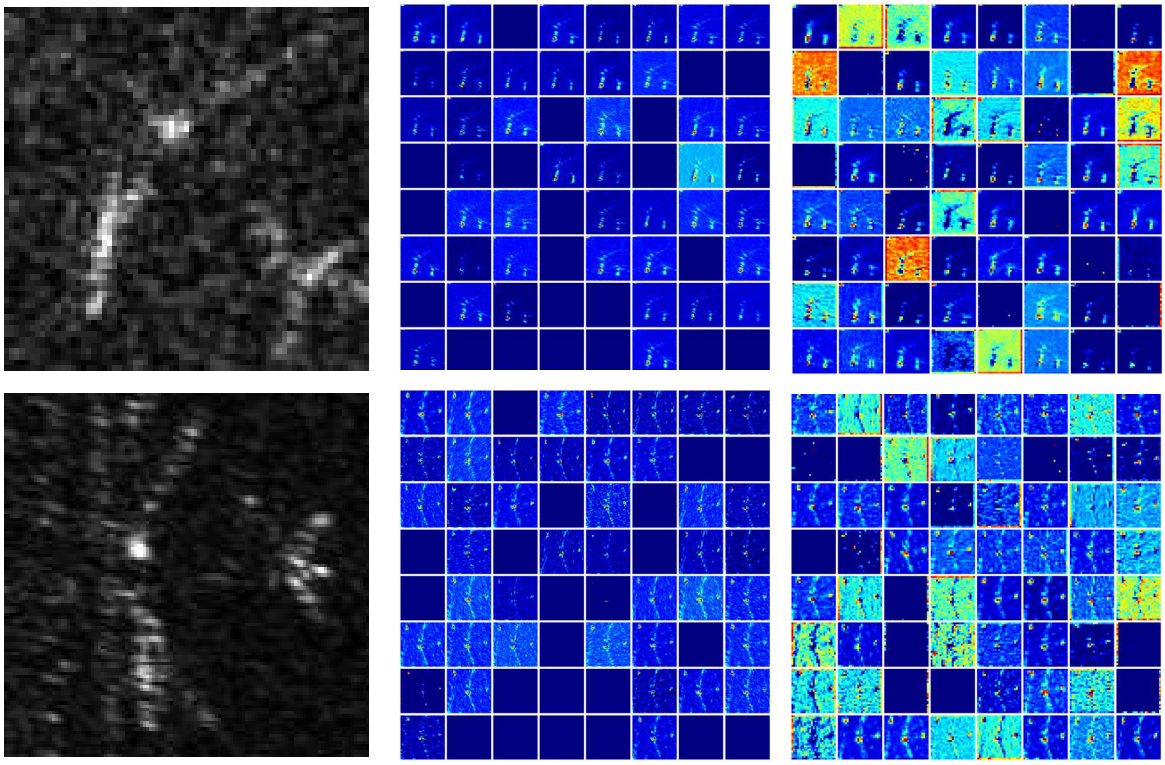


Fig. 13. Visualization of all the channels in the two shared convolution layers of the model with SEM and FEM (**Experiment I**). The first row is Boeing787, and the second row is A220.

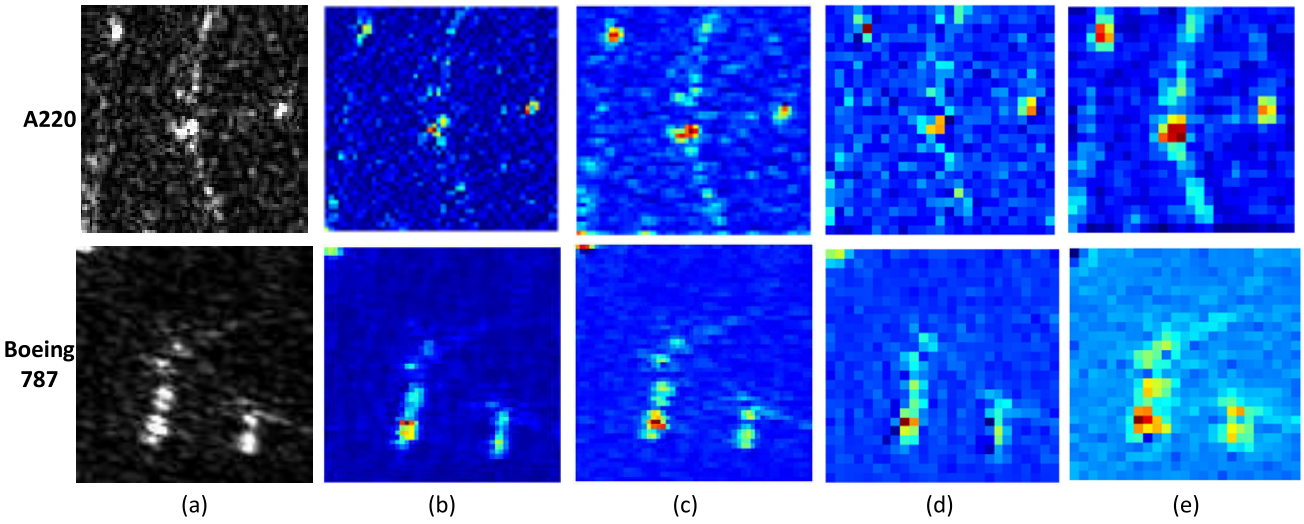


Fig. 14. Feature map visualization comparison of the baseline model and the model with SEM and FEM (**Experiment I**). (a) Illustrates the original input images. (b) Illustrates the visualization of the first convolutional layer of the baseline model. (c) Illustrates the visualization of the first convolutional layer of the model with SEM and FEM. (d) Illustrates the visualization of the second convolutional layer of the baseline model. (e) Illustrates the visualization of the second convolutional layer of the model with SEM and FEM.

stage is also conducted to prove the availability of the pretraining stage. The comparison of the test results is shown in Tables II and III. For both the 5-shot and 1-shot cases, the test accuracy is improved to a certain extent. The experimental result also reflects that pretraining fully excavates the prior information of the existing data.

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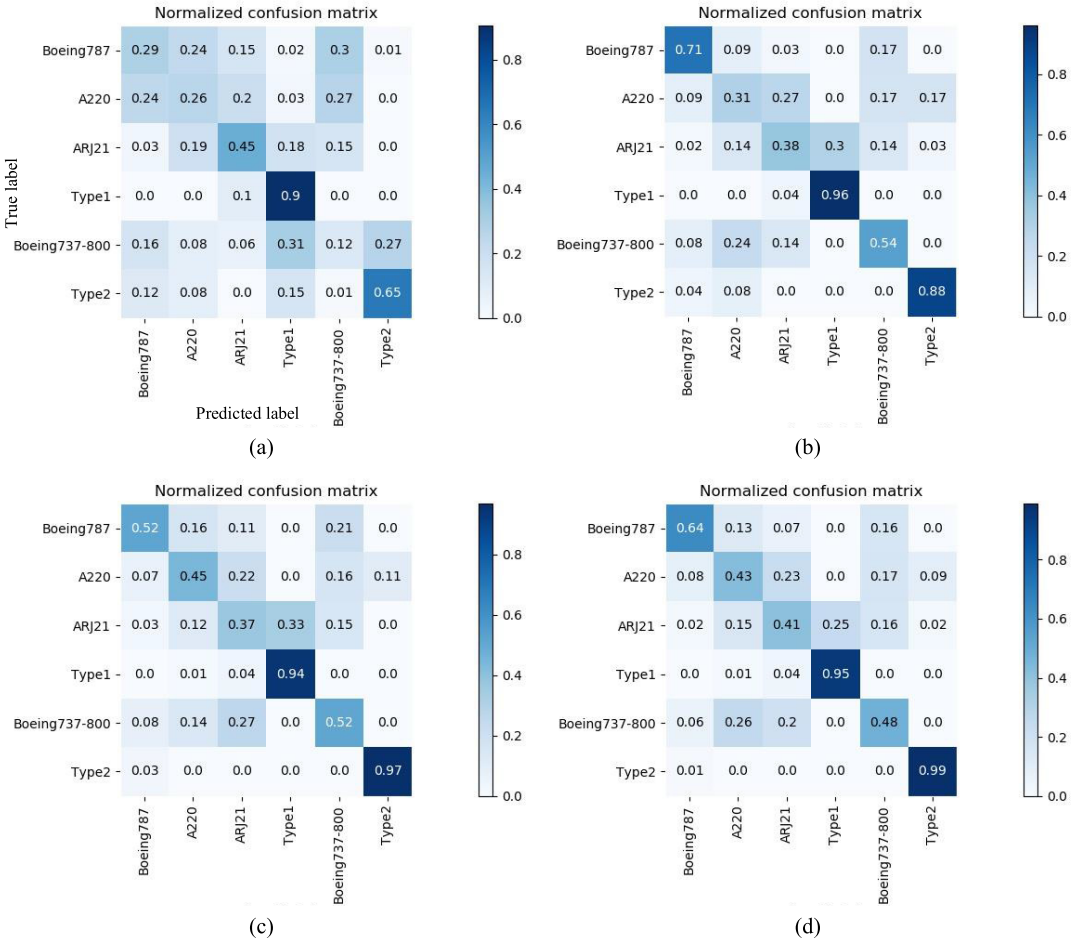


Fig. 15. Visualization of the classification results by confusion matrixes. The matrixes show the test results of the metatesting dataset with different modules for 5-shot (**Experiment I**). (a) Confusion matrix of the baseline network. (b) Confusion matrix with FEM. (c) Confusion matrix with SEM. (d) Confusion matrix with both FEM and SEM.

TABLE IX
COMPARISON WITH OTHER FSL MODELS

Methods	5-way 1-shot(%)	5-way 5-shot(%)
kNN	22.50	36.30
RelationNet	56.05 ± 0.80	66.71 ± 0.70
MAML [70]	52.2 ± 0.62	67.5 ± 0.52
Matching network	54.54 ± 0.63	58.9 ± 0.50
MSAR [39]	56.57 ± 0.62	65.15 ± 0.50
SCAN	**57.92± 0.60**	**71.84± 0.52**

*D. **Experiment I** on ASC*

The effectiveness of ASC is further illustrated by contrasting the results with or without this module. What needs illustration is that the weight of classification loss per positive sample pair is determined by the angle difference, while, in the 5-way 5-shot condition, the center representation vector is obtained by averaging five support samples, and the average angle value is meaningless. As a consequence, only the comparative experiment on the 5-way 1-shot condition is conducted. The model with AG-classifier yields good classification performance, as shown in Table IV.

*E. Comprehensive Performance Comparison Analysis of **Experiment I***

The confusion matrixes for six new test categories are visualized in Fig. 15. It indicates that the classification accuracy of "Type1" and "Type2" is relatively high. For the rest of civil aircraft, it is relatively difficult to classify accurately due to the imaging interference existing in civil airports. Finally, the classification accuracy of civil aircraft increased by at least 20%, which demonstrates the advancement of all modules. The classification accuracy curves with different modules are displayed in Fig. 16. The classification results of several test patches in this experiment are shown in Table VII. SCAN can classify the aircraft patches with different TAAs more accurately. We also test our model on large-scale SAR images. Parts of the Hongqiao Airport and Capital Airport are selected to do the slice classification experiment, and several typical sliding windows are highlighted in Fig. 17. The classification

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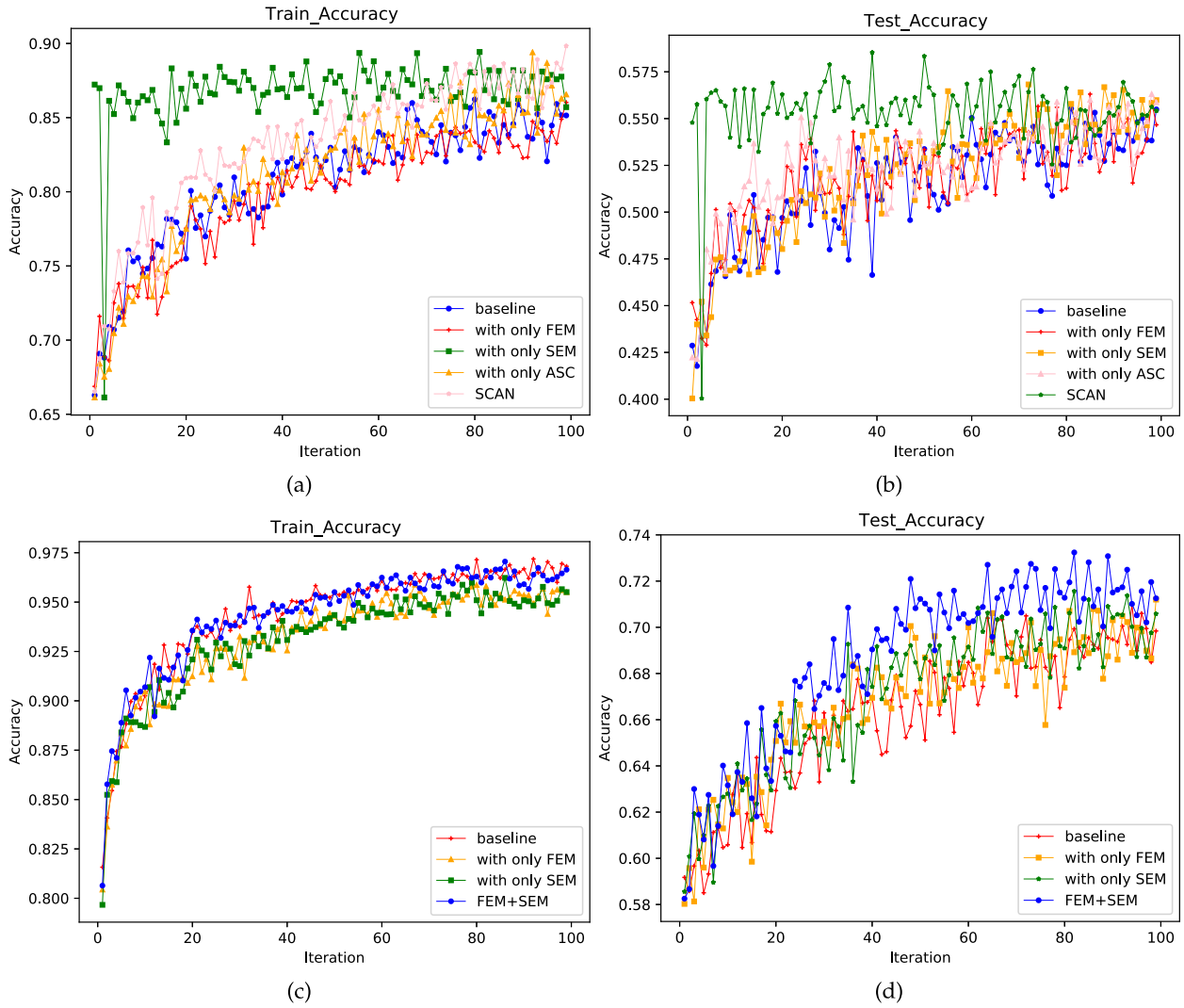


Fig. 16. Classification accuracy with different modules during training and testing (**Experiment I**). (a) Training accuracy in the 5-way 1-shot condition. (b) Testing accuracy in the 5-way 1-shot condition. (c) Training accuracy in the case of 5-way 5-shot. (d) Testing accuracy in the case of 5-way 5-shot.

TABLE X

ABLATION STUDY ON THREE RANDOMLY SELECTED METATEST SETS. THE FLUCTUATION RANGES OF THE ACCURACY OF THE THREE GROUPS OF EXPERIMENTS ARE SIMILAR. THE FLUCTUATION RANGES OF THE 5-WAY 1-SHOT TASK AND THE 5-WAY 5-SHOT TASK ARE 0.0060–0.0064 AND 0.0062–0.0065, RESPECTIVELY

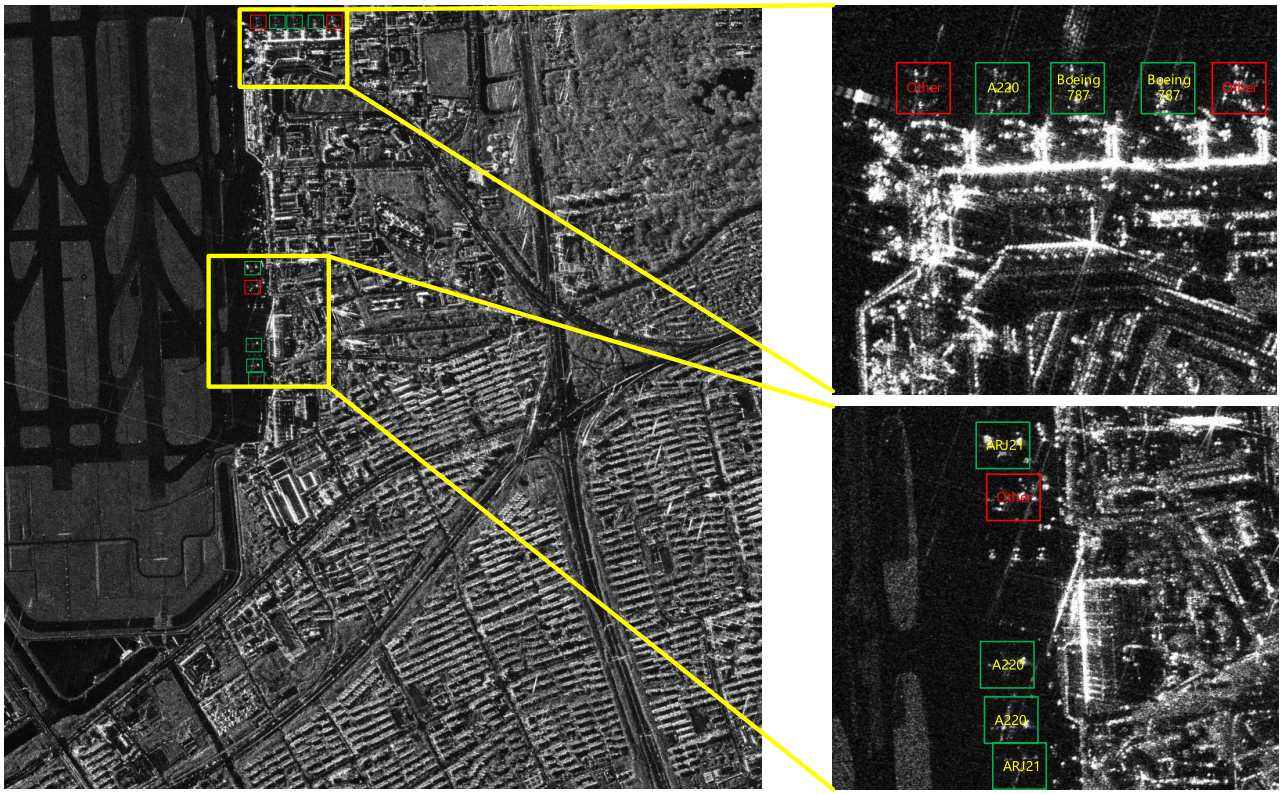
	Group1					Group2					Group3			
	SEM	ASC	FEM	Acc(%)		SEM	ASC	FEM	Acc(%)		SEM	ASC	FEM	Acc(%)
5-way 1-shot	×	×	×	53.18	5-way 1-shot	×	×	×	68.75	5-way 1-shot	×	×	×	61.94
	×	✓	✓	57.82		×	✓	✓	69.84		×	✓	✓	62.86
	✓	×	×	55.55		✓	×	×	69.24		✓	×	×	62.47
	✓	✓	✓	**57.92**		✓	✓	✓	**70.21**		✓	✓	✓	**63.83**
5-way 5-shot	×	-	×	68.66	5-way 5-shot	×	-	×	80.91	5-way 5-shot	×	-	×	74.51
	×	-	✓	69.90		×	-	✓	82.84		×	-	✓	76.33
	✓	-	×	70.04		✓	-	×	82.65		✓	-	×	76.28
	✓	-	✓	**71.04**		✓	-	✓	**83.37**		✓	-	✓	**77.13**

result verifies the stability and effectiveness of the proposed model.

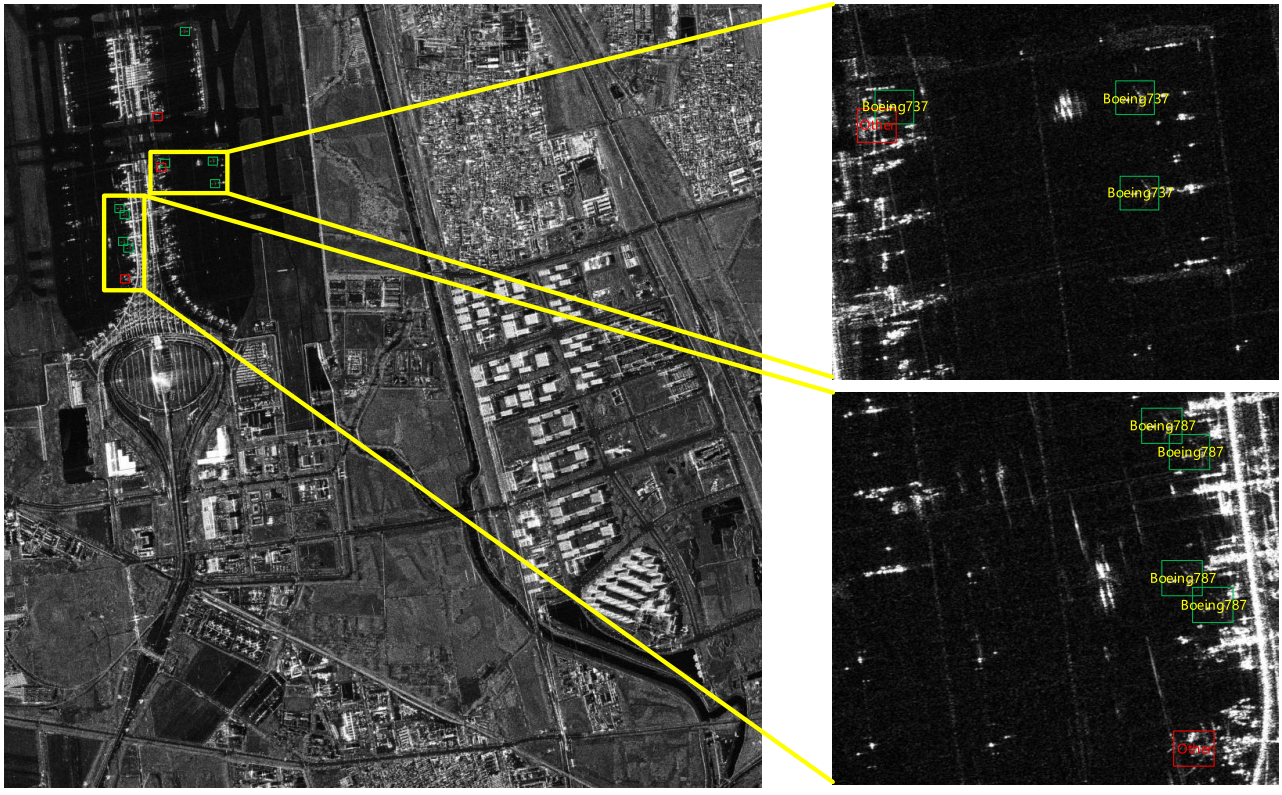
*F. 20-Class Classification Evaluation of **Experiment II***

In this section, metalearning is used for classification among all 20 categories. The percentage of metatest images is fixed at 20% for each category. Different proportions will be selected for the remaining 80% images to form the metatraining set. To validate the effectiveness of the metalearning rule, we contrast the 20-class classification results using metalearning with those classical fully connected classification networks. In the metatest phase, the 20-way 1-shot task with two query

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(a)



(b)

Fig. 17. Large scene classification results of different airports. (a) Shanghai Hongqiao International Airport. (b) Beijing Capital International Airport.

examples is taken as the basic task unit where the total number of query examples is $20 * 2 = 40$. To ensure that the experiment is as fair as possible, the batch size of the fully connected layer network equals the number of metatesting query examples 40. The detailed settings are shown in Table VI.

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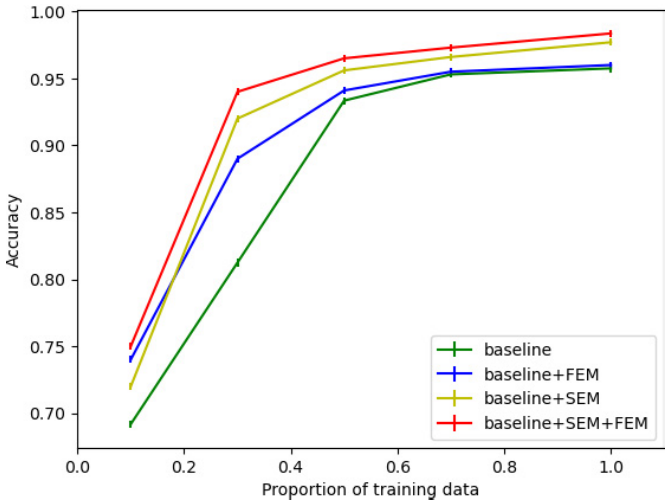


Fig. 18. Ablation study of FEM and SEM with different data proportions on the 20-way 1-shot condition. The error bar for each method fluctuates between 0.0026 and 0.0027. We can see that the fluctuation of accuracy decreases with the increase in the number of test categories (**Experiment II**).

1) Ablation Study on ASC: The effect of ASC on experiment performance is specifically studied. The classification accuracy in the 1-shot condition is shown in Table VIII. The accuracy of the metalearning classification with ASC is far more than that of a classic classification network with fully connected layers. It can achieve 90% accuracy with only 50% data. Besides, the ablation study of FEM and SEM with different proportions in **Experiment II** is also carried out. As is explained in Fig. 18, the test results verify the efficiency of both the modules.

G. Comparison With Other Methods and

As shown in Table IX, SCAN achieves the best accuracy on both 5-way 1-shot and 5-way 5-shot tasks among the compared methods. Compared with the previous state-of-the-art method [39], SCAN improves by 1.35% and 6.69% on 5-way 1-shot and 5-way 5-shot tasks, respectively. In addition, in order to fully verify the generalization ability of the proposed recognition model, we randomly select different groups of test sets for experimental verification. The experimental results are shown in Table X. The performance of the model has been effectively improved on several randomly selected metatest sets containing new categories.

V. Conclusion

In this article, a novel framework named SCAN is proposed to solve the few-shot classification problems in the SAR domain. In contrast with the existing few-shot classification models, SCAN efficiently integrates the scattering characteristics. First, SEM is proposed to combine the target imaging mechanism with the network. Second, ASC is designed to solve the problem caused by the sensitivity of the SAR imaging results to TAAs. Third, we leverage the proposed FEM to exploit frequency-domain information. Experiments on SAR-ACD demonstrate the validation of the proposed method. The model performance has increased by 4.8% for 5-way 1-shot after adding these modules. Compared with other methods with similar structures, our method achieves state-of-the-art performance for 5-way 5-shot.

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